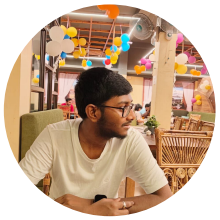
Kusal Sarkar

B.Tech ECE (2024–2028) — Jalpaiguri Government Engineering College
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GitHub1 — GitHub2 — LinkedIn



Profile

ECE undergraduate with strong interest in signal processing, embedded systems, and full stack development. Experienced in MATLAB-based signal analysis, Arduino projects, and building web applications using React and Django. Seeking internship opportunities in software and electronics roles.

Education

Jalpaiguri Government Engineering College 2024 – 2028
B.Tech in Electronics and Communication Engineering, YGPA: 8.1(2nd Year)

Bongaon High School 2021
Higher Secondary (WB) — 93.2%

Bongaon High School 2019
Secondary (WB) — 92.57%

Technical Skills

Programming: C, Java, Python
Web: HTML, CSS, JavaScript, React, Django, REST APIs, Mysql
Embedded: Arduino, Sensor Interfacing, Serial Communication
Tools: Git, MATLAB, LTSpice, Wireshark, Vivado, Figma, Linux
Core ECE: Signals and Systems, Analog/Digital Electronics, Microprocessors, Communication Systems

Projects

Signal Filtering using MATLAB

- Implemented FIR and IIR filters; analyzed signals using FFT.
- Applied sampling, reconstruction, and noise reduction techniques.

Arduino Sensor Interface System

- Built sensor-based modules with serial communication and data logging.
- Developed small automation workflows using microcontrollers.

Full Stack E-commerce Website (React + Django)

- Built responsive frontend and REST APIs for product and order handling.

Full Stack Project – Vidhya Rakshak

- Developed a prototype web application to enhance educational resource accessibility using data visualization and AI insights that will help to detect drop-out chances. Implemented authentication, analytics dashboards, and secure backend APIs using MERN stack.

Portfolio Website

- Designed and deployed a personal portfolio showcasing technical projects and achievements built using React.

Voice Agent

- Designed and deployed a personal Voice Agent for Local E-commerce

Certifications

- Meta — Full Stack Developer
- Python for Everybody — University of Michigan
- NPTEL — Electric and Magnetic Circuits (IIT Kharagpur)
- NPTEL — Mobile Virtual Reality and AI (IIT Mandi)
- IIT DHARWAD — National Capacity Building Program on Research and Publications
- Ex Jadavpur University intern - Micro-electronics Technology and VLSI Design

Language Proficiency

English , Hindi , Bengali

Soft Skills

Problem Solving, Teamwork, Adaptability, Time Management, Self-Learning